IEEE WIE ILS 2026

National Hackathon

Empower to Inspire: Thriving Together for an Inclusive and Sustainable Future

Submission Deadline: 25th August 2026

Submit as a single PDF, 12 pages max (including all sections, tables, and declaration)

Team Name HACKBROS	**Track TRACK 3: SHELEADS**
Problem Statement # / Title : **5. Safe Route Navigator** **Build an app that suggests the safest routes based on crime data, crowd density, lighting and reports**	**Solution Type:** **☐ Hardware ☑Software ☐ Hybrid**

What's Included
Do not alter this template's structure, section order, or formatting. Fill in only the marked fields/boxes. Total submission must not exceed 12 pages, including all tables, prompts, and the declaration page — submissions over 12 pages will not be evaluated. Be precise and to the point within each word-count budget.
Team Information
1. Problem Understanding
2. Proposed Solution / Idea
3. Innovation & Uniqueness
4. Feasibility & Implementation Plan
5. Expected Impact
6. Technology Stack
7. Team Member Roles
Originality & Declaration

Judging Criteria (Round 1)

Criterion (Weight)	What judges look for
Problem Understanding (10%)	Clarity, evidence, relevance to inclusion/sustainability
Solution & Innovation (25%)	Originality vs. existing solutions; how directly it solves the stated problem
Feasibility & Implementation (20%)	Realistic plan, timeline, resource awareness, risk mitigation
Expected Impact (20%)	Reach, measurability, alignment with inclusion/sustainability goals
Technology & Execution (15%)	Appropriate tech/hardware choices, prototype/demo maturity
Team & Presentation (10%)	Clear roles, well-organized submission, originality declaration complete

Team Information

Institution / Organization	Amrita Vishwa Vidhyapeetham
City / State	Thiruvallur/ Tamil Nadu
Team Lead Name	V. Suraj
Email	surajvenkatesh29@gmail.com
Phone	6374808659

Team Members

Name	IEEE Member ID	Email	Phone
Vaishnav Aditya	nil	adityakalagatoori9@gmail.com	8125229064
Preenithi Varshana	nil	padmashan68@gmail.com	9025949293
Dharani	nil	dharanibalamurugan2008@gmail.com	97913 51737

Name	IEEE Member ID	Email	Phone
Shamitha	nil	shamithavelugu@gmail.com	80150 99552
Chiru Hasini	nil	chiruhasinigangolu@gmail.com	9121777826

Round 1 — Idea / Solution

1. Problem Understanding

- ***Unsafe route selection:*** *Existing navigation systems mainly optimize for distance, time and traffic, without treating safety as a primary routing factor.*
- ***Static safety information:*** *Existing safety platforms often rely on historical crime data, periodic audits or crowdsourced reports, which may not reflect changing local conditions.*
- ***Changing risk:*** *Lighting, crowd density, local activity and safety conditions can vary significantly depending on the* ***time and location****.*
- ***Data limitations:*** *Official crime data may be delayed or incomplete, while community reports can suffer from inconsistent coverage and verification.*
- ***Last-mile safety gap:*** *In ride-hailing, safety support often ends when the vehicle reaches the destination, even though the user may still have to walk through an unfamiliar or poorly lit area.*
- ***Target users:*** *Women, students, solo travellers, families, night-time commuters and other users who want additional safety information while travelling.*

- ***Why it matters:*** *Inclusive mobility requires users to have access to **better-informed and context-aware travel choices**, rather than simply the fastest route.*

2. Proposed Solution / Idea

ABHAYA — Dynamic, Risk-Aware Navigation

- ***Dynamic Risk Engine:*** *Combines crime history, infrastructure, crowd/activity levels, recent reports, nearby safety points and time of travel to calculate comparative risk for road segments.*
- ***Personalized Safety Profiles:*** *Users can select profiles such as **Women/Solo Traveller, Family or Night Traveller**, allowing different safety factors to receive different priorities.*
- ***Risk-Aware Routing:*** *Represents roads as a weighted graph and calculates a practical route using safety, distance and travel-time costs rather than choosing only the shortest route.*
- ***Computer Vision Safety Audit:*** *A lightweight CV model can analyze available street imagery to identify indicators such as streetlights, visibility barriers and open establishments.*
- ***Police & Safe Points:*** *Displays nearby police stations and verified safety locations where reliable information is available.*
- ***Community Reports:*** *Allows users to report conditions such as broken lighting, isolated areas or recent safety concerns, while distinguishing verified information from unverified reports.*
- ***Time-Dependent Risk:*** *Risk values are adjusted according to the expected travel time so that the same road can have different risk levels at different times.*

Uber Safety Integration

- ***SafeDrop:*** *Suggests a comparatively safer nearby pickup/drop-off location when the exact destination point has higher identified risk.*
- ***RouteGuard:*** *Compares the expected ride route with observed movement and provides a discreet safety check when a significant deviation occurs.*

- ***WalkGuard:*** *After the ride ends, provides an optional short-duration safety mode for the user's final walking segment.*
- ***SOS Workflow:*** *Allows the user to activate emergency location sharing with selected contacts and, where authorized integrations exist, emergency services.*

System Flow

Crime + Infrastructure + Crowd + Reports + Time

↓

Dynamic Risk Engine

↓

Risk-Weighted Road Graph

↓

Personalized Route

↓

SafeDrop → RouteGuard → WalkGuard / SOS

3. Innovation & Uniqueness

How ABHAYA differs

- ***Dynamic instead of static:*** *Risk changes according to time, location and available evidence.*
- ***Multi-source risk:*** *Combines crime, infrastructure, activity, community reports and geographical context instead of relying on one safety indicator.*
- ***Personalized safety:*** *Different users can prioritize different safety factors.*
- ***Safety-aware routing:*** *Safety becomes part of the route cost alongside distance and travel time.*
- ***Computer Vision layer:*** *Reduces dependence on purely manual infrastructure audits by automatically identifying visual safety indicators.*
- ***End-to-end journey protection:*** *Extends beyond route selection into **pickup → ride → drop-off → final walking segment**.*

Existing Solutions vs ABHAYA

Existing Solution	*Limitation*	*ABHAYA Improvement*
MySafetiPin / Safetipin	*Safety audits and crowdsourced information may become outdated*	*Adds crime data, time-dependent risk and route-level optimization*
Crime-based SafeRoute systems	*Primarily depend on historical spatial crime data*	*Combines crime with infrastructure, activity, reports and user profiles*
Conventional navigation apps	*Primarily optimize time/distance*	*Adds safety as a routing objective*
Ride-hailing systems	*Journey safety largely focuses on the vehicle trip*	*Adds SafeDrop, RouteGuard and WalkGuard*

Plan & Impact

4. Feasibility & Implementation Plan

Phase 1 — Data & Map Layer

- *Collect available government crime datasets.*
- *Integrate road-network and geographical data.*
- *Store spatial data using* ***PostGIS****.*
- *Divide the road network into connected road segments.*

Phase 2 — Dynamic Risk Engine

- *Normalize crime, lighting, crowd, report and infrastructure indicators.*
- *Apply time-dependent weighting.*
- *Generate* ***Low / Moderate / High*** *comparative risk levels.*
- *Maintain timestamps and confidence levels for data sources.*

Phase 3 — Risk-Aware Routing

- *Represent intersections as **vertices** and roads as **edges**.*
- *Assign dynamic costs to road segments.*
- *Implement **Dijkstra / A***.*
- *Balance safety against excessive distance and travel time.*

Phase 4 — Computer Vision

- *Train/use a lightweight CV model on suitable street imagery.*
- *Detect infrastructure indicators such as:*
    - *Streetlights*
    - *Visibility barriers*
    - *Open establishments*
- *Feed detected conditions into the risk engine.*

Phase 5 — Uber Integration

- *Implement **SafeDrop** using nearby location-risk analysis.*
- *Implement **RouteGuard** using expected vs observed route information.*
- *Implement **WalkGuard** as a post-ride safety workflow.*
- *Use simulated/API-based data where production API access is unavailable.*

Phase 6 — Validation

- *Test multiple routes under different times and risk conditions.*
- *Compare shortest-route and safety-aware-route results.*
- *Measure route-generation time and risk reduction.*
- *Conduct usability testing before wider deployment.*

Key Constraints & Mitigation

- ***Incomplete crime data** → show data coverage/confidence instead of assuming missing data means zero risk.*
- ***Outdated information** → timestamp data and periodically recalculate affected areas.*
- ***False community reports** → distinguish verified and unverified reports.*
- ***Privacy concerns** → use explicit consent and minimize location-data retention.*

- ***Third-party API limitations*** *→ use simulated data for prototype demonstrations and integrate official APIs when authorized.*

5. Expected Impact

Primary Beneficiaries

- *Women travelling alone*
- *Students and young commuters*
- *Solo travellers*
- *Families*
- *Night-time commuters*
- *Ride-hailing users*
- *Communities and local authorities*

Expected Benefits

- ***Better route decisions:*** *Users receive safety context in addition to travel time.*
- ***Improved awareness:*** *Users can identify changing risk conditions before and during travel.*
- ***Safer ride-hailing experience:*** *SafeDrop addresses unsafe pickup/drop-off locations.*
- ***Last-mile support:*** *WalkGuard extends safety awareness beyond the end of the vehicle journey.*
- ***Community participation:*** *User reports can highlight emerging infrastructure or safety concerns.*
- ***Infrastructure insights:*** *Repeated reports can help identify areas requiring attention.*

Measurable KPIs

- ***Risk reduction:*** *Difference in calculated risk between conventional and ABHAYA routes.*
- ***Travel-time overhead:*** *Additional time required to obtain a lower-risk route.*
- ***Route computation time:*** *Time taken to generate a recommendation.*
- ***CV performance:*** *Accuracy/precision of infrastructure detection.*

- ***User usability:*** *User ratings for clarity and usefulness.*
- ***SafeDrop suitability:*** *Percentage of recommended drop points satisfying predefined safety criteria.*
- ***Data freshness:*** *Percentage of risk inputs with valid timestamps.*

Inclusion & Sustainability

- *Supports **inclusive mobility** by allowing different users to choose safety priorities relevant to their circumstances.*
- *Can improve confidence in walking and shared mobility rather than requiring users to avoid travel because of safety uncertainty.*
- *A software-first architecture allows expansion to additional cities without requiring dedicated physical infrastructure.*

Technology & Team

6. Technology Stack

Layer	*Technologies*
Frontend	*React.js, Tailwind CSS*
Maps	*Leaflet, OpenStreetMap*
Backend	*Python, FastAPI*
Database	*PostgreSQL, PostGIS*
Data Processing	*Pandas, NumPy*
AI/ML	*Scikit-learn, CNN / Object Detection*
Risk Engine	*Python-based weighted risk model*
Route Optimization	*Graphs, Dijkstra, A**
Geospatial Processing	*PostGIS, GeoJSON*
Ride Integration	*Uber APIs / SDKs where available*
Emergency Workflow	*Location APIs, consent-based live sharing*
Deployment	*Docker, Cloud*

***Hardware** Not Applicable — **Software-only** prototype.*

AI/ML & Open Source

- *ML: Used for risk modelling and infrastructure assessment.*
- *Computer Vision: Used to identify visual infrastructure indicators.*
- *Graph Algorithms: Used for explainable route optimization.*
- *Open-source components: Used for mapping, geospatial processing and software development to improve feasibility and reduce cost.*

User Validation

☐ Yes ☑ No

Formal user/field research has not yet been conducted. Initial validation will use scenario-based testing followed by structured user testing.

7. Team Member Roles

List only as many rows as your team size (1–6). Solo participants: complete just one row.

Member Name	Role	Responsibilities
Member 1	AI/ML LEAD	Crime-data preprocessing, dynamic risk model, risk scoring and ML evaluation.
Member 2	Computer Vision Lead	Street-level image analysis, streetlight/visibility detection and CV integration.
Member 3	DSA And Routing Lead	Road graph construction, dynamic edge weighting, Dijkstra/A* and route optimization.

Member Name	Role	Responsibilities
Member 4	Backend And Database Lead	FastAPI, PostgreSQL/PostGIS, data pipelines and backend APIs.
Member 5	Frontend And UX Lead	React interface, interactive maps, safety visualization, profiles and user experience.
Member 6	Uber and Safety Integration Lead	SafeDrop, RouteGuard, WalkGuard, SOS and emergency-contact workflow.

Originality & Declaration

We, the undersigned team, declare that:

☑ The idea/solution submitted is original and has been conceived and developed by our team.

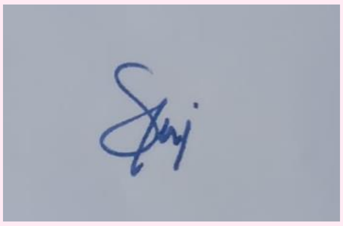
☑ This submission (or a substantially similar version) has not won a prior hackathon or competition award.

☑ Any third-party tools, datasets, libraries, images, or references used have been duly credited within this document.

☑ We understand that plagiarism, fabrication of data, or misrepresentation of authorship will lead to disqualification from IEEE WIE ILS 2026.

Team Lead Name & Signature (or e-signature):

V. Suraj



Date: 24/08/2026

Questions? Contact Madhav Maheshwari (+919588874415) or wieils2026@gmail.com